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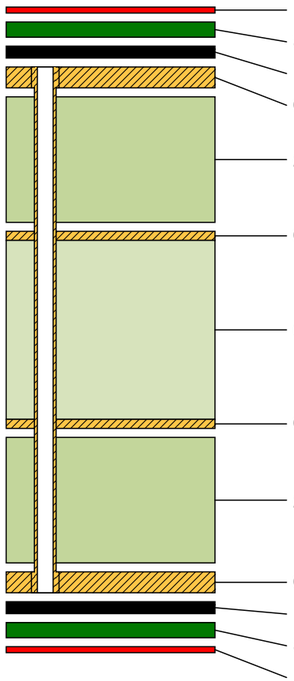
REVISION	DESCRIPTION	DATE	APPROVED

Layer Stack Legend



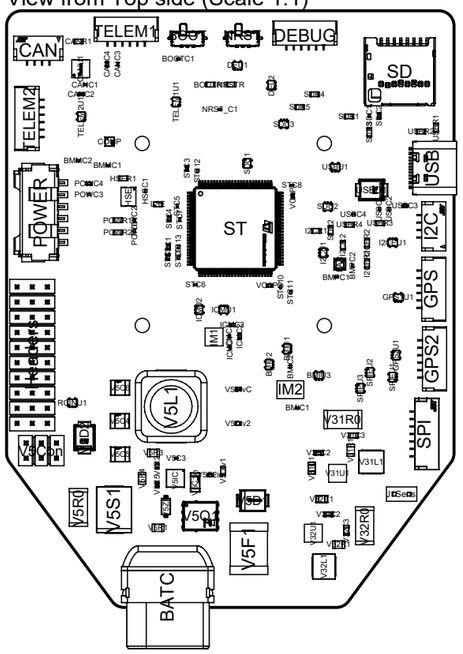
Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.03mm	SM-001	Solder Mask	GTS
PbSn	Top Surface Finish	0.02mm		Surface Finish	
CF-004	**Top Layer**	**0.04mm**		**Signal**	**GTL**
Prepreg		*0.21mm*	*7628*	*Dielectric*	
CF-003	**Int1 (GND)**	**0.02mm**		**Signal**	**G1**
Core		*1.06mm*	*FR4*	*Dielectric*	
CF-003	**Int2 (PWR)**	**0.02mm**		**Signal**	**G2**
Prepreg		*0.21mm*	*7628*	*Dielectric*	
CF-004	**Bottom Layer**	**0.04mm**		**Signal**	**GBL**
PbSn	Bottom Surface Finish	0.02mm		Surface Finish	
Surface Material	Bottom Solder	0.03mm	SM-001	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 1.68mm

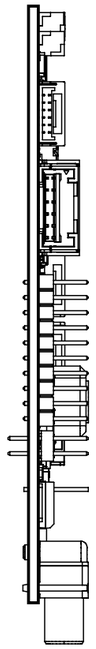
Bill Of Materials

Line #	Designator	Comment	Quantity
1	BAR	BMP388	1
2	BATC	XT60PW-F	1
3	BMIC1, BMIC2, BMMC1, BMMC2, BMPC1, BMPC2, BOOTC1, CANC1, CANC3, HSEC1, ICMC1, ICMC4, NRST_C1, POWC1, POWC2, POWC3, SDC1, STC5, STC6, STC7, STC8, STC9, STC10, STC11, STC12, STC13, USBC1, USBC2, USBC4, V5C3	100nF 1mOhm	30
4	BMIU1, BMIU2, BMIU3, DEU1, DEU2, GPS1U1, GPS2U1, I2CEU1, I2CU1, ICMU1, ICMU2, RCINU1, SDU1, SDU2, SDU3, SPIEU1, SPIEU2, SPIEU3, TELEM1U1, TELEM2U1, USBU1	TPD2E2U06DRLR	21
5	BOOT_SW, NRST_SW	SSAJ120100	2
6	BOOTR1, NRSTR, SDR1, SDR2, SDR3, SDR4, SDR5, USBR4	10kOhm	8
7	CAN, I2C_EXT	SM04B-GHS-TB(LF)(SN)	2
8	CANC2, CANC4, ICMC2, USBC3, VCAP1, VCAP2	2.2uF	6
9	CANR1	120Ohm	1
10	CANU1	TJA1051TK/3/1J	1
11	COMP	BMM150	1
12	DEBUG, GPS, GPS2, TELEM1, TELEM2	SM06B-GHS-TB(LF)(SN)	5
13	Headers	HTSW-110-07-L-T	1
14	HSE	ASEMB-24.000MHZ-LC-T	1
15	HSER1, I2CER1, I2CER2, I2CR1, I2CR2, POWR1, POWR2	2.2kOhm	7
16	ICMC3	10nF	1
17	IM1	ICM-20602	1
18	IM2	BMI088	1
19	J1Sens	68000-102HLF	1
20	LSE	SIT1552AI-JE-DCC-32.768D	1
21	POWC4, SDC2, STC1, STC2	1uF	4
22	POWER	502494-0670	1
23	SD	104031-0811	1
24	SPI_EXT	SM07B-GHS-TB(LF)(SN)	1
25	ST	STM32H743VIT6	1
26	STC3, STC4	4.7uF	2
27	USB	TYPE-C-31-M-12	1
28	USBD1, V5D2	SMBJ5.0A	2
29	USBR1, USBR2, USBR3	5.1kOhm	3
30	V5C1in	4.7uF 3mOhm	1
31	V5C2in, V5DivC	100nF 39mOhm??	2
32	V5C4, V5C5, V5C6	47uF 3mOhm	3
33	V5Con	TSW-103-08-L-D	1
34	V5D1	SMBJ36A	1
35	V5Div1	360kOhm 100mW	1
36	V5Div2, V5R2	30.9kOhm 63mW, 30.9kOhm	2
37	V5F1	3425L300_36MR	1
38	V5IC1	LMR51450FNDRRR	1
39	V5L1	10uH 16mOhm	1
40	V5Q1	SI7489DP-T1-E3	1
41	V5R0, V31R0, V32R0	AC2512JK-070RL	3
42	V5R1, V5R3, V5R4, V31R1, V32R1	100kOhm	5
43	V5R5	16.9kOhm	1
44	V5S1	SS34C	1
45	V5Z1	BZT52C15-E3-08	1
46	V31C1, V32C1	10uF 3mOhm	2
47	V31C2, V32C2	390pF 1mOhm	2
48	V31C3, V32C3	22uF 1mOhm	2
49	V31L1, V32L1	2.2uH	2
50	V31U1, V32U1	TPS62142RGTT	2

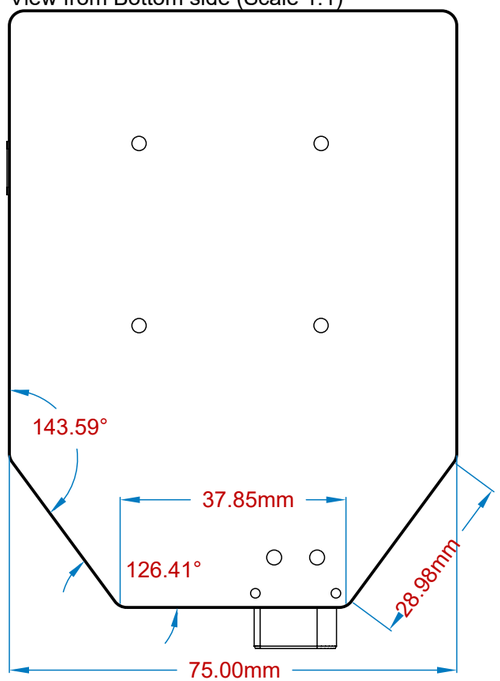
View from Top side (Scale 1:1)



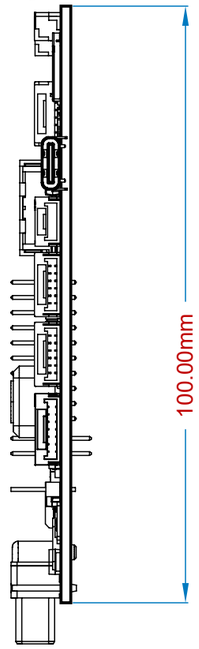
View from Left side (Scale 1:1)



View from Bottom side (Scale 1:1)



View from Right side (Scale 1:1)



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SCALE: =SCH_SCALE | FILENAME: PCB_mounting.PCBDwf | SHEET 1 OF 2

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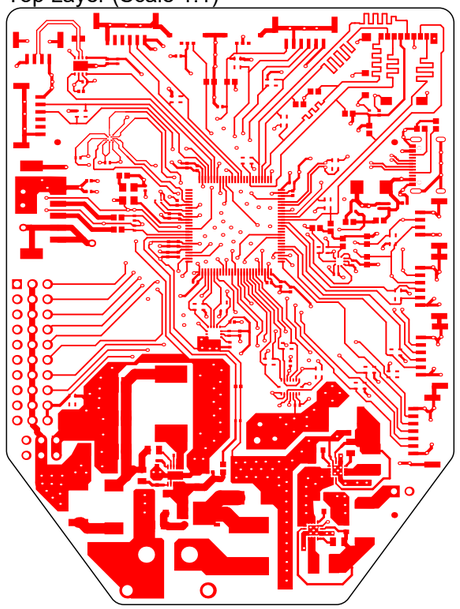
REVISION	DESCRIPTION	DATE	APPROVED

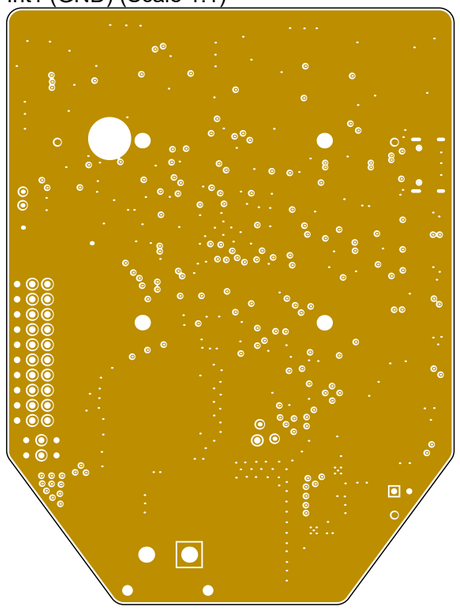
Transmission Line Structure Table

Impedance Id	Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Wide Trace Width	Narrow Trace Width	Gap	Reference layers	Substack	Clearance	Target Tolerance
1	Edge-Coupled Coated Microstrip	90	89.32	Top Layer	0.29mm	0.29mm	0.20mm	Int1 (GND)	Board Layer Stack	0.13mm	10%
2	Edge-Coupled Coated Microstrip	120	119.12	Top Layer	0.18mm	0.18mm	0.36mm	Int1 (GND)	Board Layer Stack	0.13mm	10%
3	Coated Microstrip	50	49.93	Top Layer	0.35mm	0.35mm		Int1 (GND)	Board Layer Stack	0.13mm	10%
4	Edge-Coupled Coated Microstrip	90	89.32	Bottom Layer	0.29mm	0.29mm	0.20mm	Int2 (PWR)	Board Layer Stack	0.13mm	10%
5	Edge-Coupled Coated Microstrip	120	119.12	Bottom Layer	0.18mm	0.18mm	0.36mm	Int2 (PWR)	Board Layer Stack	0.13mm	10%
6	Coated Microstrip	50	49.93	Bottom Layer	0.35mm	0.35mm		Int2 (PWR)	Board Layer Stack	0.13mm	10%

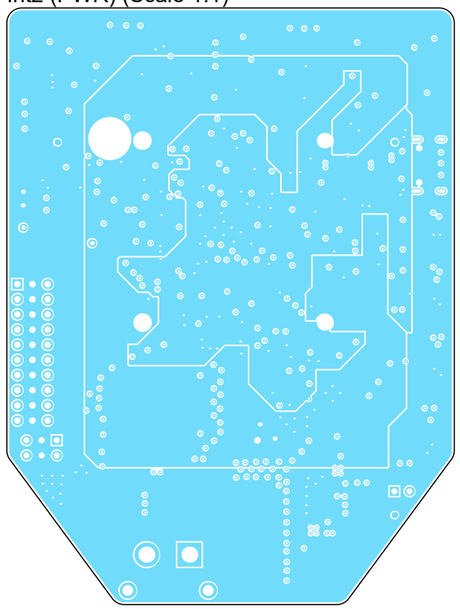
Top Layer (Scale 1:1)



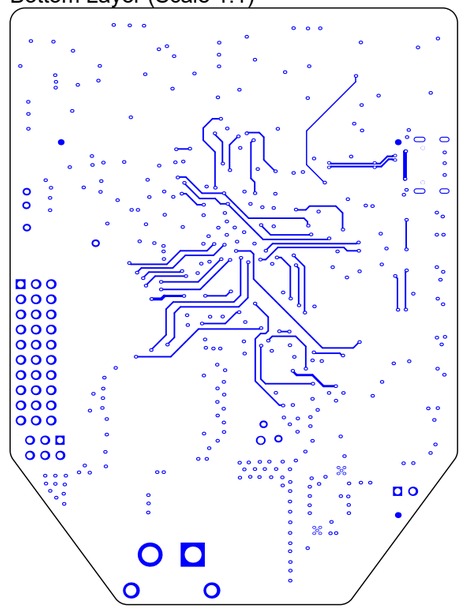
Int1 (GND) (Scale 1:1)



Int2 (PWR) (Scale 1:1)



Bottom Layer (Scale 1:1)



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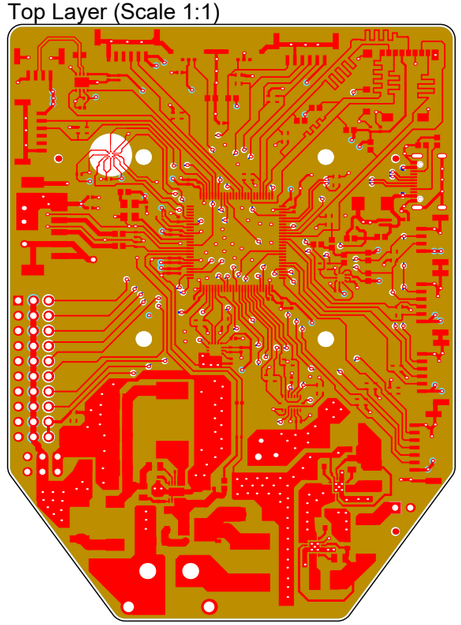
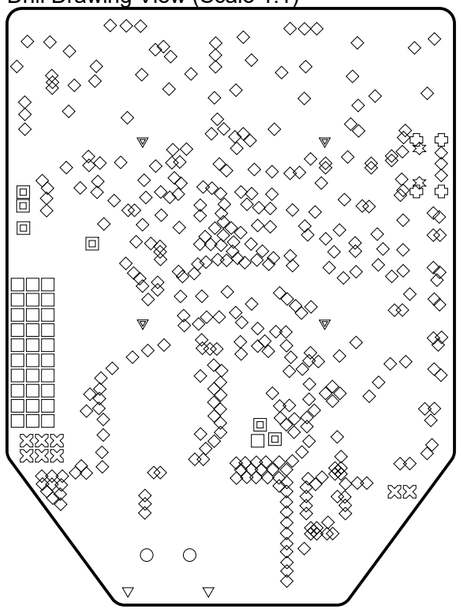


Drill Drawing View (Scale 1:1)



Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
◇	374	0.30mm	Plated	
✡	2	0.60mm	Non-Plated	
✚	4	0.60mm	Plated	
⊡	6	0.71mm	Plated	
⊠	8	1.02mm	Plated	
□	31	1.10mm	Plated	
▽	2	1.80mm	Plated	
▼	4	2.60mm	Plated	
○	2	2.70mm	Plated	
	433 Total			

APPROVALS	DATE
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Altium

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